

PCB-0026-A_gan-boost-48v

Run 20261008-094328-sim: passed, 100 of 100 steps passed.

100	0	0	32
steps passed	steps failed	steps not reached	human confirmations

Run

ITEM	VALUE
Board	PCB-0026-A_gan-boost-48v
Bench	sim
Started	2026-10-08T09:43:28Z
Finished	2026-10-08T09:43:29Z
Procedure	af3a2394a8ddce9c (generated 2026-10-08T09:43:28Z)
Firmware	2e3ab22503fcb829
netlist	29e528a3f7536cc3
constraints	5e0ab057b5021d0e
bom	55f2b91100454284
requirements	ce948868c3d5c729
manifest	57eb1867945d747f
Simulated faults	none
Seed	1

Stages

STAGE	STEPS	PASS	FAIL	NOT RUN
visual	2	2	0	0
unpowered	19	19	0	0
power-up	7	7	0	0
rails	16	16	0	0
programming	5	5	0	0
blocks	7	7	0	0
power-stage	30	30	0	0
limits	14	14	0	0

Not in this procedure: unpowered:U201 (U201 LMG2100R044 is an integrated half-bridge: no body diode to check); full-function (firmware has no spin command yet (manifest stage boost)); limits:ov (over-voltage trip at 27 V is above the rated input max 24 V; not applied).

Measurements

STEP	WHAT	LIMITS	VALUE	VERDICT
unpowered.03	VIN to GND is not a short	≥ 1000 ohm	7015.46 ohm	pass
unpowered.05	+48V to GND is not a short	≥ 1000 ohm	17711.5 ohm	pass
unpowered.07	ISNS_N to GND is not a short	≥ 1000 ohm	16456.6 ohm	pass
unpowered.09	ISNS_P to GND is not a short	≥ 1000 ohm	8826.04 ohm	pass
unpowered.11	+5V to GND is not a short	≥ 100 ohm	1243.15 ohm	pass
unpowered.13	VCC to GND is not a short	≥ 100 ohm	1174.24 ohm	pass
unpowered.15	+3V3 to GND is not a short	≥ 100 ohm	1477.39 ohm	pass
unpowered.17	VDDA to GND is not a short	≥ 100 ohm	1683.09 ohm	pass
unpowered.19	SW to GND is not a short	≥ 1000 ohm	6407.89 ohm	pass
power-up.06	idle input current	0.002 .. 0.18 A	0.0283128 A	pass
rails.02	+48V voltage	11 .. 14.14 V	13.907 V	pass
rails.04	ISNS_N voltage	13.3 .. 14.14 V	13.8506 V	pass
rails.06	ISNS_P voltage	13.3 .. 14.14 V	13.8967 V	pass
rails.08	+5V voltage	4.75 .. 5.25 V	4.92532 V	pass
rails.10	VCC voltage	4.75 .. 5.25 V	4.99181 V	pass
rails.12	+3V3 voltage	3.135 .. 3.465 V	3.32193 V	pass
rails.14	VDDA voltage	3.135 .. 3.465 V	3.27315 V	pass
rails.16	+5V ripple	≤ 0.1 V	0.0383581 V	pass

STEP	WHAT	LIMITS	VALUE	VERDICT
programming.02	flash			pass
programming.04	boot banner		fwe PCB-0026-A sim	pass
programming.05	firmware answers		OK { "board": "PCB-0026-A" }	pass
blocks.01	firmware test hook version		OK { "board": "PCB-0026-A" }	pass
blocks.02	firmware test hook status		OK { "sim": true, "state": "off", "outputs_on": false, "trips": [], "vbus_v": 14.000, "vin_v": 14.000 }	pass
blocks.03	firmware test hook selftest		OK { "sim": true }	pass
blocks.04	firmware test hook adc		OK { "sim": true }	pass
blocks.05	firmware reads the input voltage		vbus_v=14	pass
blocks.07	U301 sense-amp offset	1.6 .. 1.7 V	1.63576 V	pass
power-stage.02	reply to arm		OK { "state": "soft-start", "target_v": 48.0 }	pass
power-stage.04	the voltage loop regulates the output		vout_v=48.02	pass
power-stage.05	input current while regulating	0.002 .. 0.18 A	0.0414583 A	pass
power-stage.06	reply to disarm		OK { "state": "off", "outputs_on": false }	pass
power-stage.07	open loop at D0 = 0.491		OK { "state": "open", "duty": 0.491 }	pass
power-stage.10	+48V follows the duty	24.754 .. 30.255 V	27.1972 V	pass
power-stage.12	PWM_LO PWM frequency	980000 .. 1.02e+06 Hz	999433 Hz	pass
power-stage.13	PWM_LO duty	0.461 .. 0.521	0.489442	pass
power-stage.14	dead time PWM_LO falling to PWM_HI rising	8 .. 13 ns	9.43485 ns	pass
power-stage.15	reply to disarm		OK { "state": "off", "outputs_on": false }	pass
power-stage.16	reply to duty 0.491		OK { "state": "open", "duty": 0.491 }	pass
power-stage.17	ovp trips on a real over-voltage		EVT { "trip": { "new": ["ovp_hw"] } }	pass

STEP	WHAT	LIMITS	VALUE	VERDICT
power-stage.18	ovp dropped both outputs		out-puts_on=False	pass
power-stage.21	reply to disarm		OK {"state":"off","out-puts_on":false}	pass
power-stage.22	ovp clears once its comparator has fallen		OK {"trips":[]}	pass
power-stage.23	reply to duty 0.491		OK {"state":"open","duty":0.491}	pass
power-stage.25	ocp trip reported		EVT {"trip":{"new":["ocp_hw"]}}	pass
power-stage.26	ocp dropped both outputs		out-puts_on=False	pass
power-stage.29	reply to disarm		OK {"state":"off","out-puts_on":false}	pass
power-stage.30	ocp clears once its comparator has fallen		OK {"trips":[]}	pass
limits.03	idle current at 12 V	0.002 .. 0.18 A	0.0339858 A	pass
limits.06	idle current at 24 V	0.002 .. 0.18 A	0.0184929 A	pass
limits.08	reply to duty 0.491		OK {"state":"open","duty":0.491}	pass
limits.10	UV trip below 10.5 V		EVT {"trip":{"new":["vin_uv"]}}	pass
limits.12	reply to disarm		OK {"state":"off","out-puts_on":false}	pass
limits.13	reply to clear		OK {"trips":[]}	pass

STEP	ACTION	BY	AT	RESULT
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Human steps

A simulated run confirms its own human steps as sim.

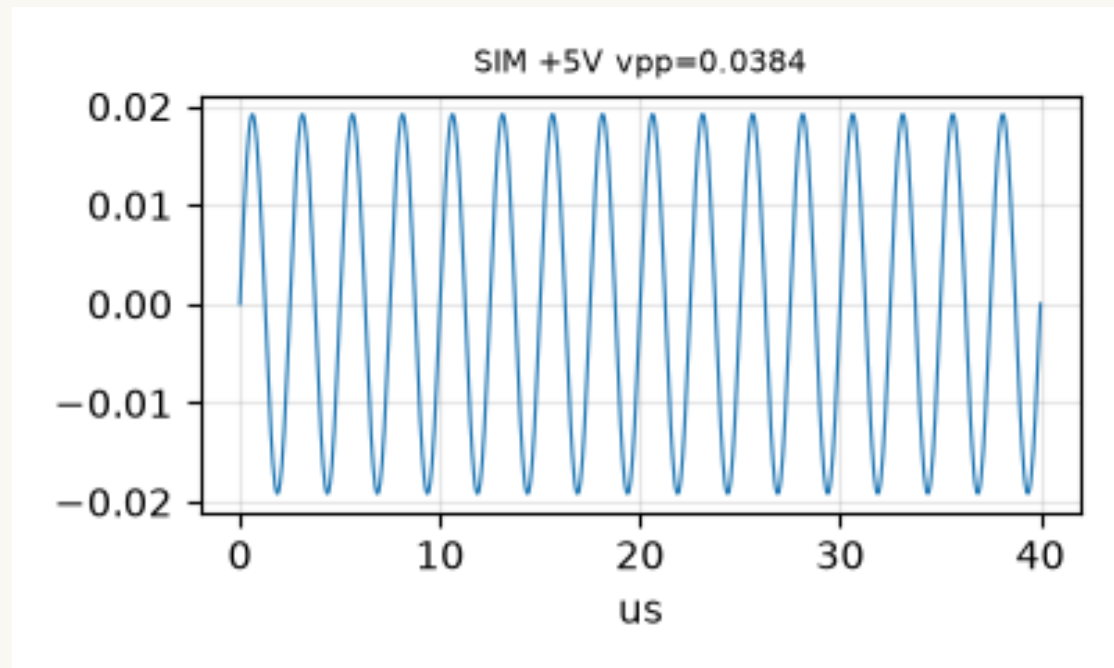
STEP	ACTION	BY	AT	RESULT
visual.01	Check the board for damage, solder bridges on fine-pitch ICs (U201, U301, U401, U402, U501) and missing parts against fab/BOM.csv.	sim	2026-10-08T09:43:28Z	pass
visual.02	Check orientation of polarised parts: D101 SMBJ28A, D201 SMBJ54A, D401 MBRX160, D501 MBRX160, U201 LMG2100R044, U301 INA240A2, U401 LMR16006X, U402 TLV75533, U501 STM32G474CBT6, Q501 CJ2310.	sim	2026-10-08T09:43:28Z	pass
unpowered.01	Board unpowered, every cable off. Discharge the input capacitors.	sim	2026-10-08T09:43:28Z	pass
unpowered.02	DMM on resistance: red (+) on TP101 (VIN), black (-) on TP102 (GND).	sim	2026-10-08T09:43:28Z	pass
unpowered.04	DMM on resistance: red (+) on TP205 (+48V), black (-) on TP102 (GND).	sim	2026-10-08T09:43:28Z	pass
unpowered.06	DMM on resistance: red (+) on NT102 pin 2 (ISNS_N), black (-) on TP102 (GND).	sim	2026-10-08T09:43:28Z	pass
unpowered.08	DMM on resistance: red (+) on NT101 pin 2 (ISNS_P), black (-) on TP102 (GND).	sim	2026-10-08T09:43:28Z	pass
unpowered.10	DMM on resistance: red (+) on TP401 (+5V), black (-) on TP102 (GND).	sim	2026-10-08T09:43:28Z	pass
unpowered.12	DMM on resistance: red (+) on C215 pin 1 (VCC), black (-) on TP102 (GND).	sim	2026-10-08T09:43:28Z	pass
unpowered.14	DMM on resistance: red (+) on TP402 (+3V3), black (-) on TP102 (GND).	sim	2026-10-08T09:43:28Z	pass
unpowered.16	DMM on resistance: red (+) on C408 pin 1 (VDDA), black (-) on TP102 (GND).	sim	2026-10-08T09:43:28Z	pass
unpowered.18	DMM on resistance: red (+) on TP201 (SW), black (-) on TP102 (GND).	sim	2026-10-08T09:43:28Z	pass

STEP	ACTION	BY	AT	RESULT
power-up.02	Connect the supply: + to TP101 (VIN), - to TP102 (GND). Supply set to 14 V / 0.2 A, output still off.	sim	2026-10-08T09:43:28Z	pass
power-up.03	About to switch the supply on at 14 V with a 0.2 A limit. Hands clear; watch for smoke or a hot part.	sim	2026-10-08T09:43:28Z	pass
power-up.07	Touch-test: no part is hot a few seconds after power-up.	sim	2026-10-08T09:43:28Z	pass
rails.01	DMM on DC volts: red (+) on TP205 (+48V), black (-) on TP102 (GND).	sim	2026-10-08T09:43:28Z	pass
rails.03	DMM on DC volts: red (+) on NT102 pin 2 (ISNS_N), black (-) on TP102 (GND).	sim	2026-10-08T09:43:28Z	pass
rails.05	DMM on DC volts: red (+) on NT101 pin 2 (ISNS_P), black (-) on TP102 (GND).	sim	2026-10-08T09:43:28Z	pass
rails.07	DMM on DC volts: red (+) on TP401 (+5V), black (-) on TP102 (GND).	sim	2026-10-08T09:43:28Z	pass
rails.09	DMM on DC volts: red (+) on C215 pin 1 (VCC), black (-) on TP102 (GND).	sim	2026-10-08T09:43:28Z	pass
rails.11	DMM on DC volts: red (+) on TP402 (+3V3), black (-) on TP102 (GND).	sim	2026-10-08T09:43:28Z	pass
rails.13	DMM on DC volts: red (+) on C408 pin 1 (VDDA), black (-) on TP102 (GND).	sim	2026-10-08T09:43:28Z	pass
rails.15	Scope channel 1 on TP401 (+5V), ground spring to the nearest GND pad, AC coupled, 20 MHz bandwidth limit.	sim	2026-10-08T09:43:28Z	pass
programming.01	Connect the SWD probe to J501; the board stays powered from the supply.	sim	2026-10-08T09:43:29Z	pass
programming.03	Connect the USB-UART (3V3, 115200 8N1) to J502: adapter RX to pin 1 (board TX), adapter TX to pin 2 (board RX), GND to GND. Then press reset.	sim	2026-10-08T09:43:29Z	pass
blocks.06	DMM on DC volts: red (+) on R301 pin 1 (ISNS_RAW), black (-) on TP102 (GND).	sim	2026-10-08T09:43:29Z	pass
power-stage.01	Output open: nothing on +48V's connectors. The next steps switch U201 at 1 MHz from the 14 V supply; the output reaches 55 V in the over-voltage test. Hands clear.	sim	2026-10-08T09:43:29Z	pass
power-stage.09	DMM on DC volts: red (+) on TP205 (+48V), black (-) on TP102 (GND).	sim	2026-10-08T09:43:29Z	pass

STEP	ACTION	BY	AT	RESULT
power-stage.11	Scope channel 1 on TP203 (PWM_LO) and channel 2 on TP202 (PWM_HI), 10x probes with ground springs to the nearest GND pad, DC coupled, full bandwidth (at least 200 MHz for the dead time).	sim	2026-10-08T09:43:29Z	pass
power-stage.19	+48V stays charged near 55 V and bleeds down through its divider; do not touch it.	sim	2026-10-08T09:43:29Z	pass
power-stage.24	Inject on TP301 (ISNS): a second supply at 0 V with a 20 mA limit, + through 100 ohm to TP301 (ISNS), - to TP102 (GND). Raise it slowly until the console shows the ocp trip; stop at 3.3 V if it does not.	sim	2026-10-08T09:43:29Z	pass
power-stage.27	Turn the injection to 0 V and disconnect it from TP301 (ISNS).	sim	2026-10-08T09:43:29Z	pass

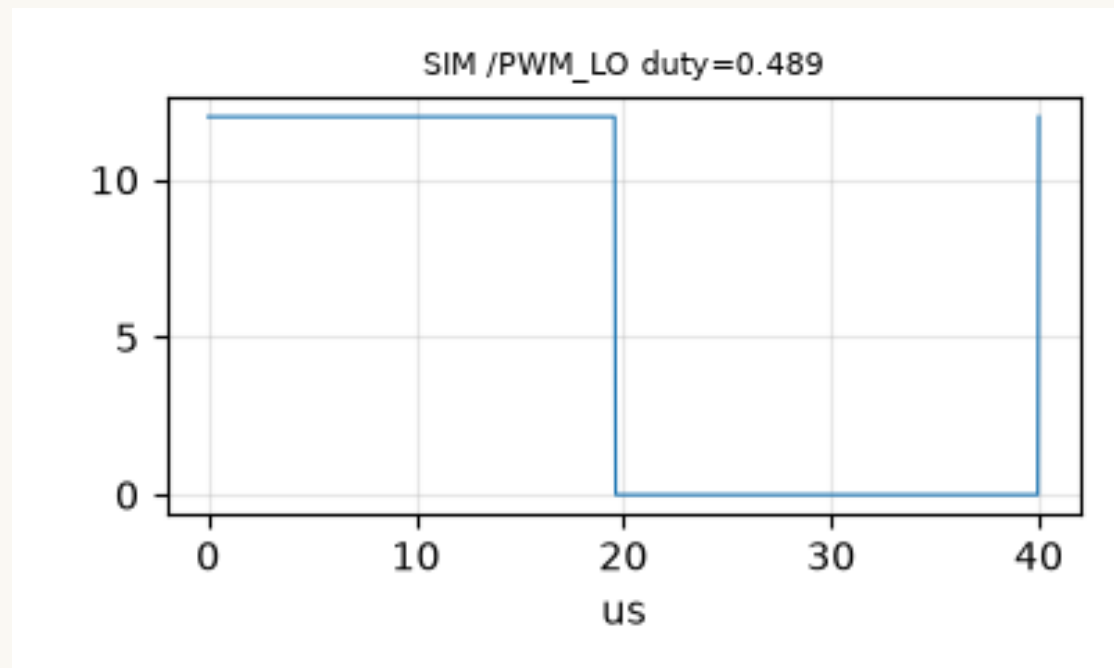
Scope captures

RAILS.16



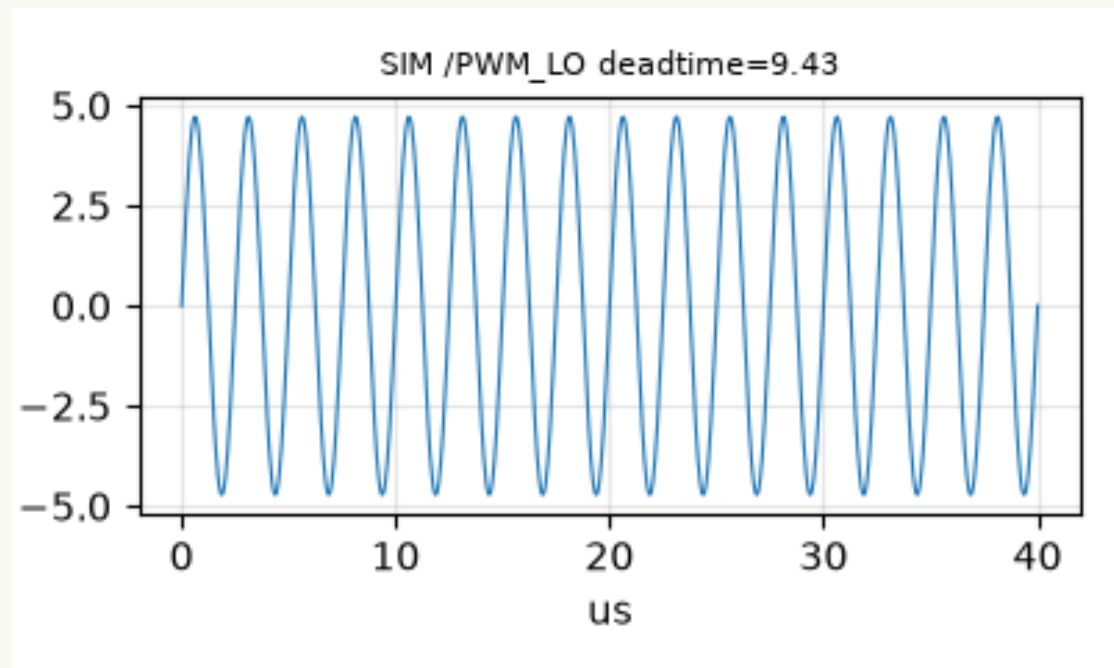
+5V ripple (captures/rails.16.png)

POWER-STAGE.13



PWM_LO duty (captures/power-stage.13.png)

POWER-STAGE.14



dead time PWM_LO falling to PWM_HI rising (captures/power-stage.14.png)